VLA Continuum and Molecular Line Observations of Jets from High-Mass Protostars

Ananay Sethi[1], Spencer J. Ireland[1], Peter Hofner[1,2], Tatiana M. Rodríguez[3], Esteban D. Araya[4,1]

[1]New Mexico Tech [2]National Radio Astronomy Observatory [3]University of Cologne [4]Western Illinois University

Introduction

High-Mass stars ($M \geq 8M_\odot$) are crucial in determining the physical and chemical dynamic processes in the Galaxy. During their proto-stellar phase these stars drive ionized jets and molecular outflows which play a fundamental role in the formation of a high-mass star. At New Mexico Tech, the High-Mass Star Formation Group conducts high-resolution, high-sensitivity radio continuum and molecular line observations of high-mass star-forming regions. In this contribution we present early results on two prominent regions: 1) The southern source IRAS 16562-3959 (left) for which we have obtained VLA-A array observations in the S, C, X and Ku continuum bands which fully resolve the structure of the ionized jet, and 2) VLA A-array spectral line data of IRAS 18566+0408 (right) in NH_3 and CH_3OH transitions. Both studies are presently ongoing and are examples of our current research activities.

Protostellar Jet IRAS 16562-3959

Observations:

- VLA S (3 GHz), C (6 GHz), bands continuum observations in A-configuration.
- Synthesized beam: $2''.09 \times 0''.51$ (3 GHz), $1''.06 \times 0''.26$ (6 GHz). RMS: 37μJy/beam (3 GHz), 8.75μJy/beam (6 GHz).

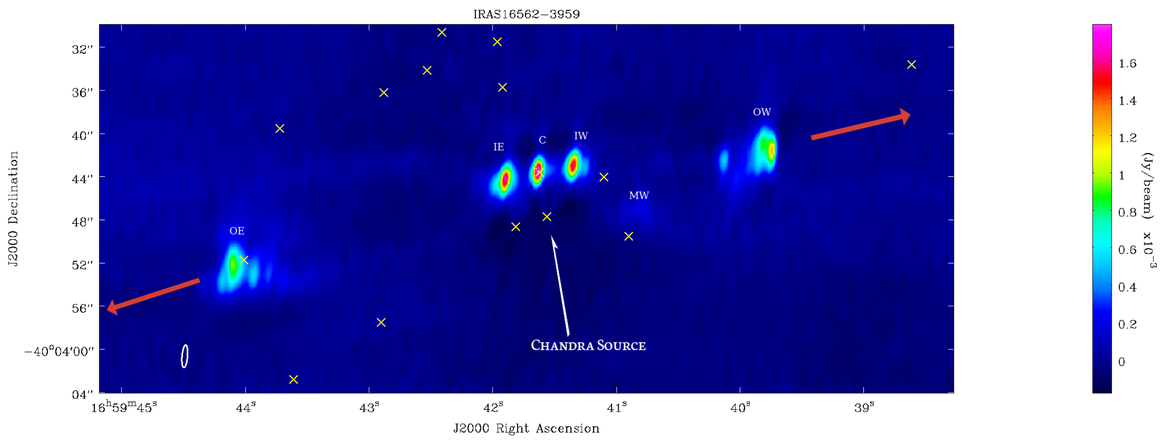

Figure 1. Closeup of IRAS 16562-3959 at S-Band (3 GHz), with X-ray detections from the Montes et al. (2020) observations using Chandra (shown with 'X' symbols). Separate regions of the protostellar jet have been labeled using nomenclature inspired by Guzman et al. (2010), moving East-to-West - Outer-East (OE), Inner-East (IE), Central (C), Inner-West (IW), Middle-West (MW), Outer-West (OW). The red arrows portray the projected direction of the jets from the central core.

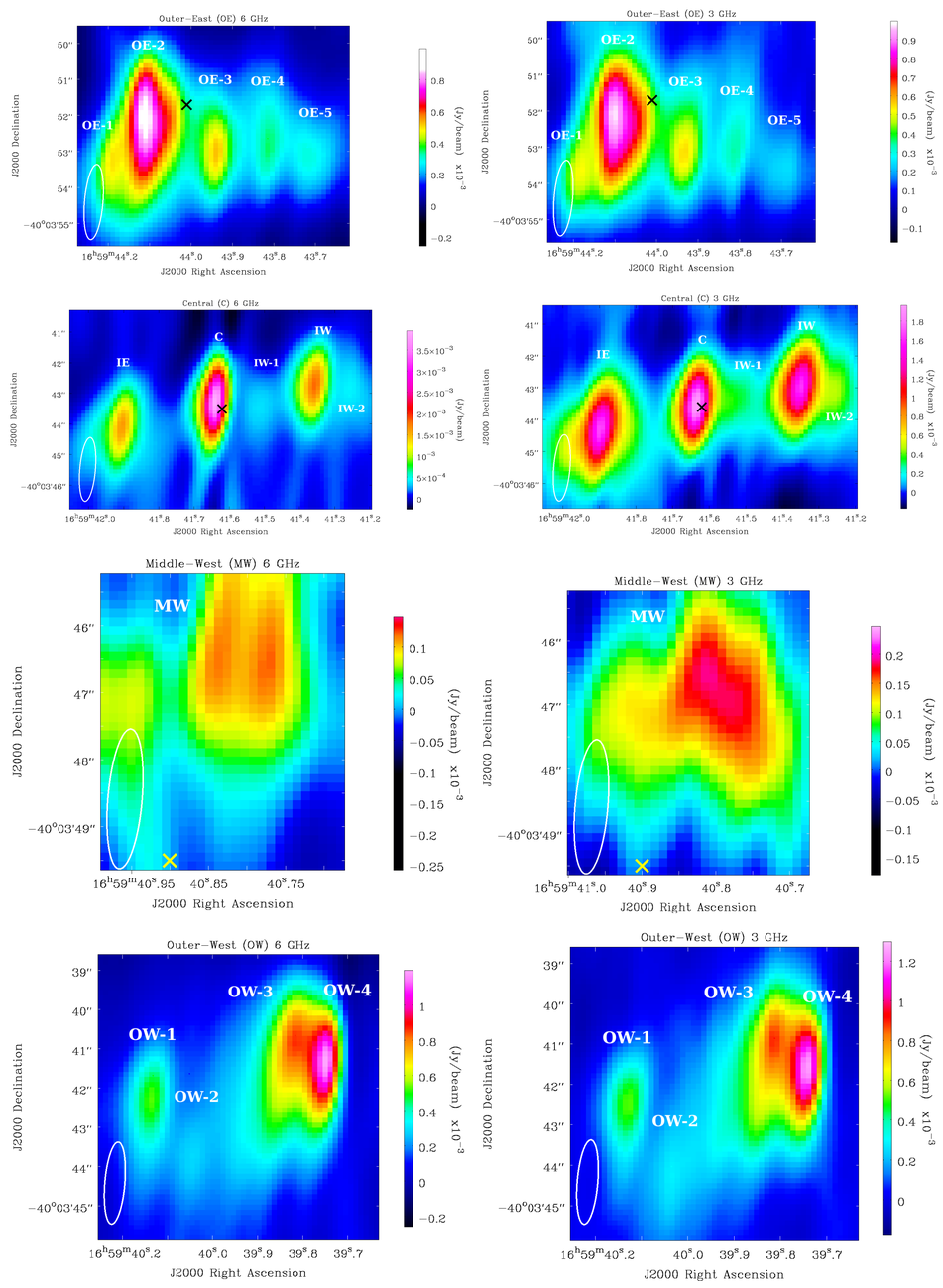

Figure 2. Comparison of regions across 6 GHz (left) and 3 GHz (right) maps. The 6 GHz images were smoothed and re-gridded to reach the same resolution as the 3 GHz images, as can be seen with the beam size on both the left and right. Due to higher resolution and sensitivity, individual components were detected in these regions, and are numerically categorized to maintain nomenclature.

Preliminary Results:

- We present the highest sensitivity and resolution observations yet toward this ionized jet.
- The jet from the central core travels along the North-West/South-East direction. Both outer regions seems to be offset to the south from the main trajectory of the jet as defined by the inner region.
- The Central source C is thermal, with a spectral index of $\alpha_{C,S} = 1.0 \pm 0.001$.
- We report the detection of new individual components within the IW, OE, OW regions of the jet, and we also report a new detection of a region of synchrotron emission called Middle-West (MW), with a spectral index of $\alpha_{C,S} = -0.6 \pm 0.002$.
- X-ray detections seem to be closely associated to Sources C, MW, and OE - warranting further study of shock dynamics and low-mass protstellar flares in these regions.

Ionized Jet Candidate IRAS 18566+0408

Observations:

- VLA K-band (18-26.5 GHz) continuum and spectral line observations in A-configuration.
- 1024 channels with 0.18 km s^{-1} spectral resolution.
- Synthesized beam: $0''.12 \times 0''.09$
- RMS: 1.98 mJy beam^{-1}
- Target spectral lines: NH_3 (3,3) and (6,6) inversion transitions and CH_3OH $J_k = 7_2 - 7_1$ and $J_k = 9_2 - 9_1$ transitions.
- Extended emission imaged by applying a $0''.3$ uvtaper and decreasing the spectral resolution to 1.0 km s^{-1}.

Maps and Spectra:

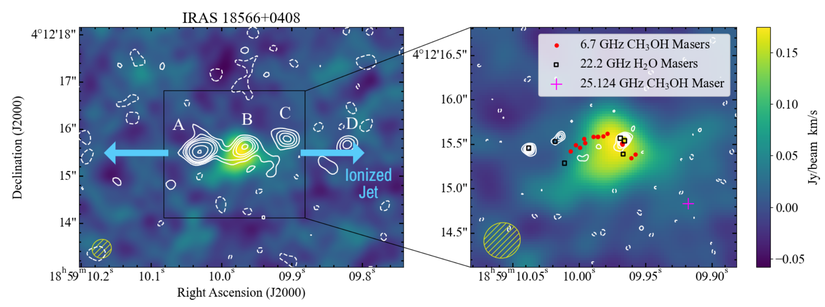

Figure 3. NH_3 (3,3) inversion transition integrated intensity map, including emission from the hyperfine structure main and satellite lines. *Left:* The white contours show the 6 cm continuum of Rosero et al. (2016). *Right:* A closer look at the integrated intensity map with the 1.3 cm continuum from Rodríguez et al. (in prep) overlaid as white contours.

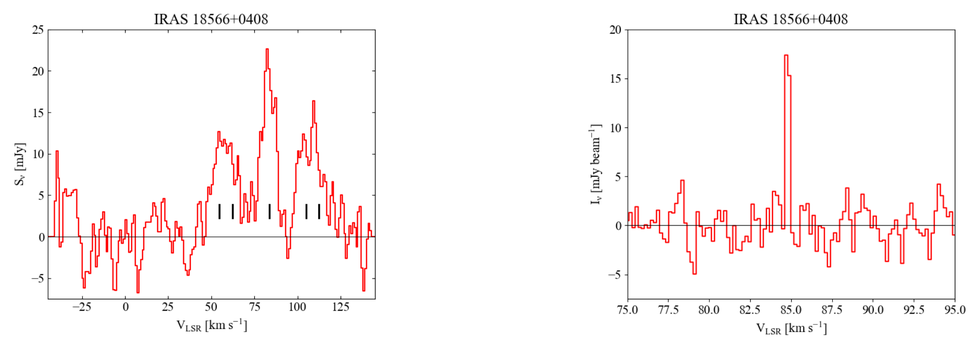

Figure 4. *Left:* NH_3 (3,3) inversion transition spectrum containing emission within a region defined as 3σ above the integrated intensity map's RMS noise level. The bold lines represent the positions of the hyperfine structure main and satellite components. *Right*: Peak position spectrum of the 25.124 GHz CH_3OH maser.

Preliminary Results:

- NH_3 (3,3) inversion transition properties: Peak brightness temperature $T_B^{peak} = 123$ K, optical depth $\tau \approx 20$, and total column density $N_{\rm tot}(NH_3) = 6.7 \times 10^{17}$ cm^{-2}. These numbers suggest a central hydrogen density $n(H_2) \approx 10^8$ cm^{-3}.
- We find that the NH_3(3,3) peak is slightly offset from continuum component B. Presuming that the NH_3 peak indicates the position of the exciting protostar, this would favor shock ionization rather than photoionization for component B.
- We detected a compact source in the CH_3OH $J_k = 7_2 - 7_1$ transition with a peak brightness temperature $T_B^{peak} \approx 3500$ K. We conclude that the source is a newly discovered Class I 25.124 GHz CH_3OH maser that is collisionally-pumped by outflows. It is approximately $1''.3$ from the string of 6.7 GHz CH_3OH masers (Araya et al. 2010).

References & Acknowledgments

• Araya et el. 2010, ApJL, 717, L133 • Guzmán et al. 2010, ApJ, 725, 734

• Montes et al. 2020, ApJ, 888, 118 • Rodríguez et al. in prep.

• Rosero et al. 2016, ApJS, 227, 25

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Contact Information:

Ananay Sethi: ananay.sethi@student.nmt.edu

Spencer J. Ireland: spencer.ireland@student.nmt.edu